

Developer Technology Trends: *Data Analysis Report*



IBM Capstone Project



Outline

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Executive Summary

Current Technology

JavaScript & PostgreSQL lead while AWS & Google Cloud dominate platforms

Future Technology

JavaScript remains #1 language as PostgreSQL stays the most desired database. AWS leads as the most wanted platform..

Demographics

Developer base mainly aged 24–35 and largely holds a Bachelor's degree, and from USA.

Introduction

This report analyzes technology trends among developers using data subset from the Stack Overflow Developer Survey and GitHub Jobs API. It examines current and future technology preferences, along with key demographic patterns in the global developer community.

The objective is to support data driven decisions on technology investments, talent development, and hiring strategy. It is intended for developers, HR professionals, technology leaders, and organizations looking to align their skills, hiring, and training strategies with current developer trends.



Methodology

- **DATASET**

- Stack Overflow Developer Survey (subset)
 - GitHub Jobs API
-

- **DATA COLLECTION**

- GitHub Jobs API and web scraping
-

- **DATA CLEANING**

- Removal of null & duplicate values
 - Standardized data types & formats
 - Normalization of multi-value columns
-

- **DATA EXPLORATION & VISUALIZATION**

- Python (pandas, Matplotlib)
-

- **DASHBOARD CREATION**

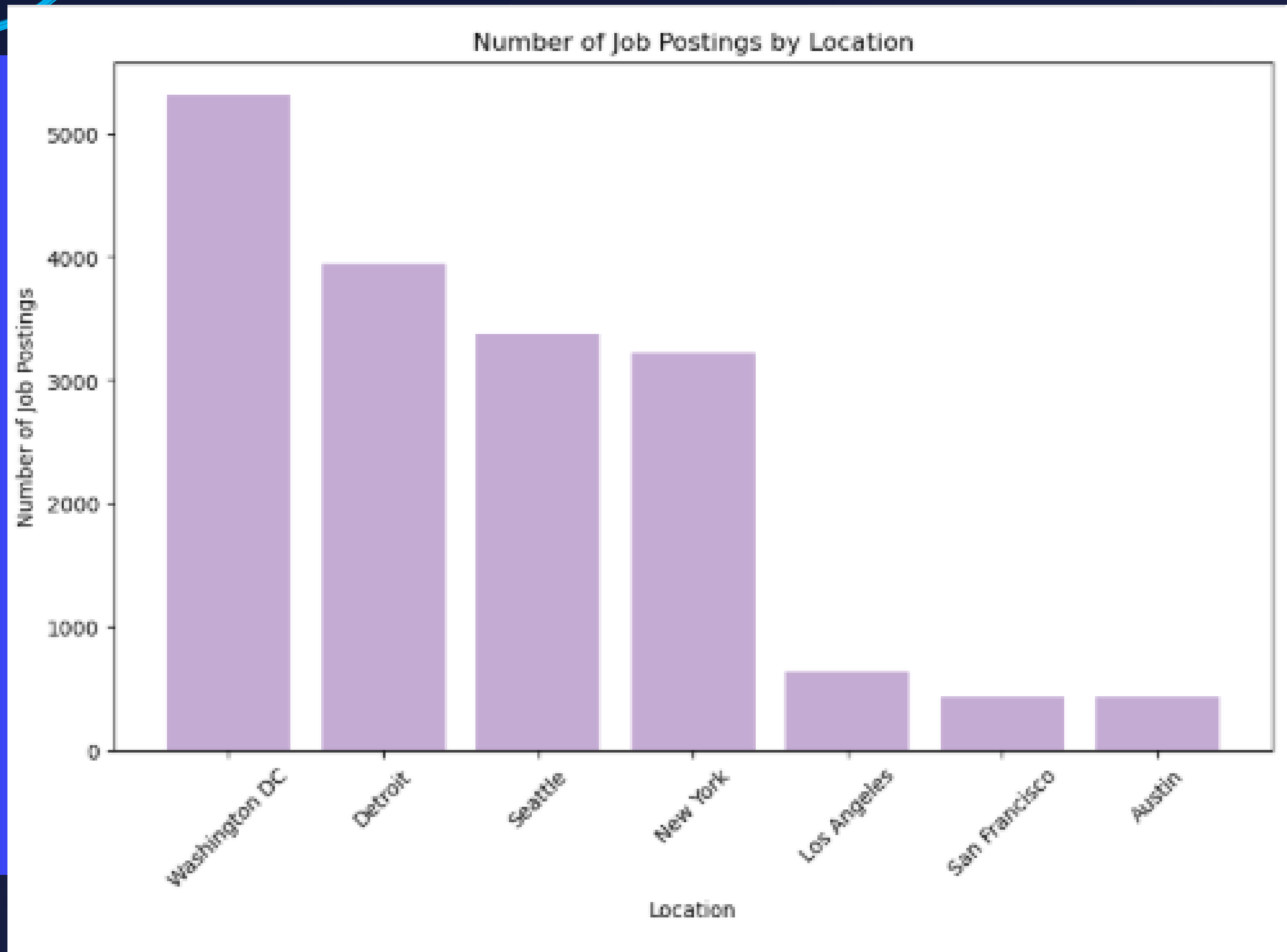
- IBM Cognos Analytics
-



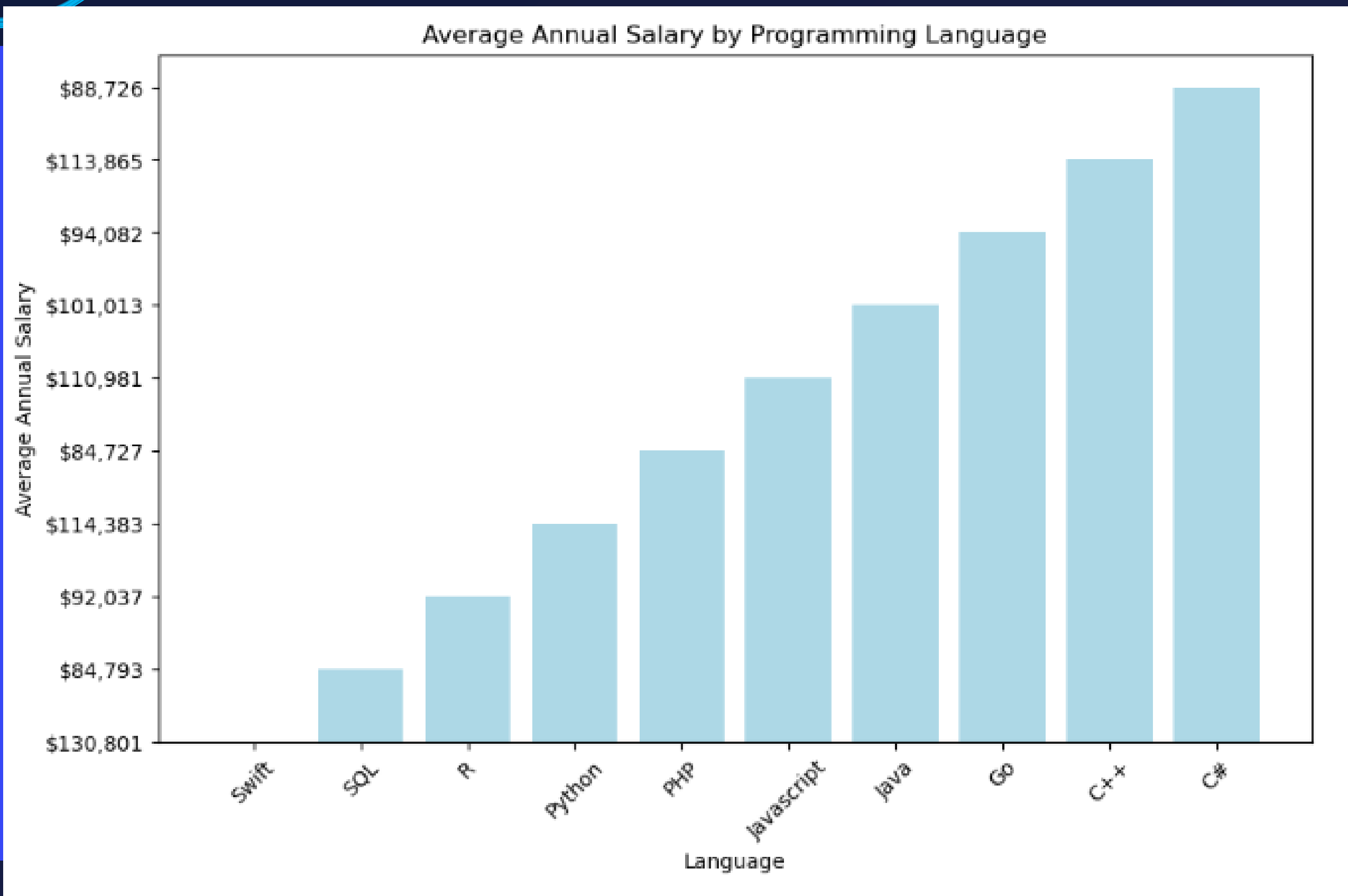
Results



Job Postings - Jobs API



Popular Languages

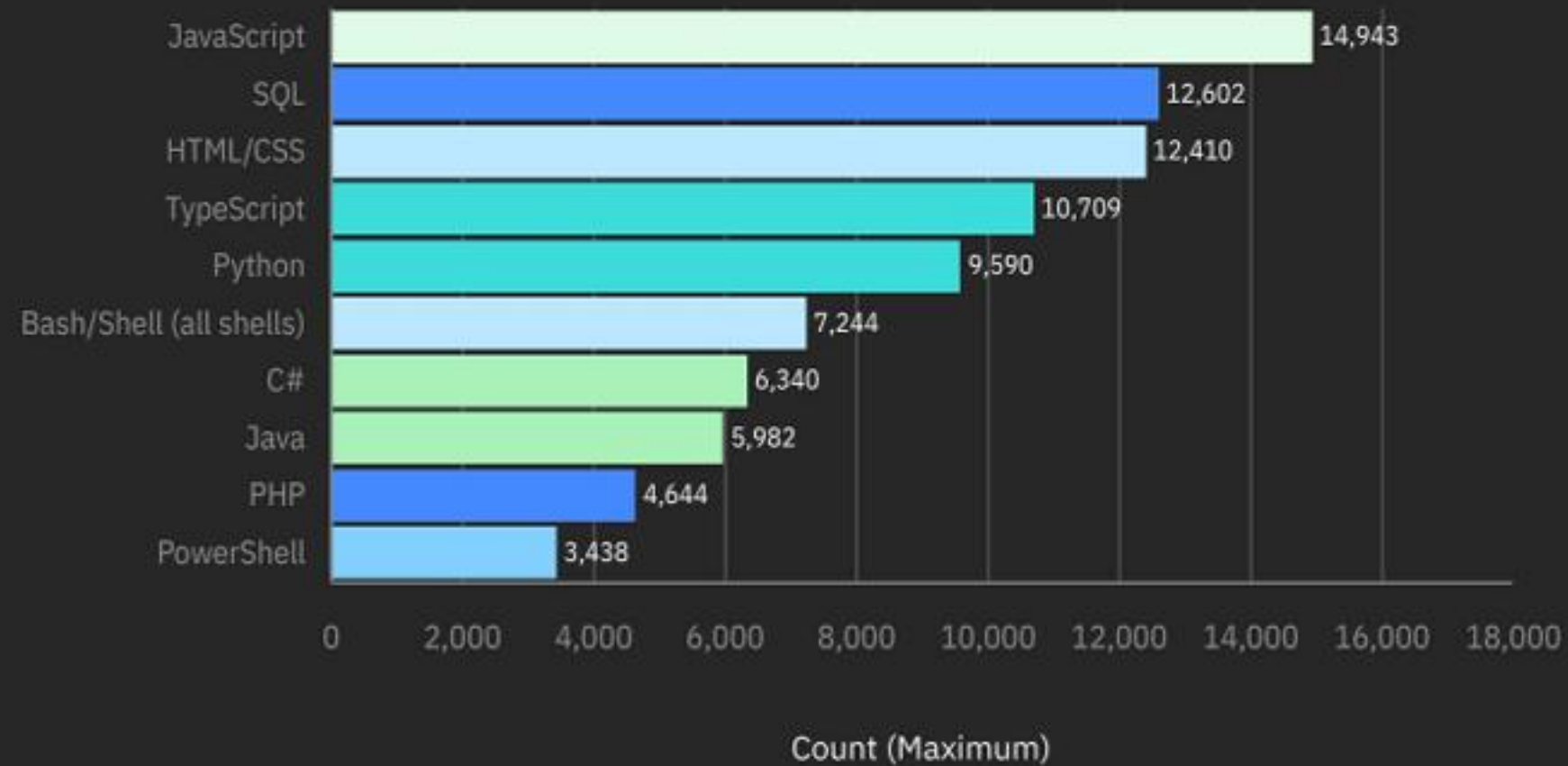


Programming Language Trends

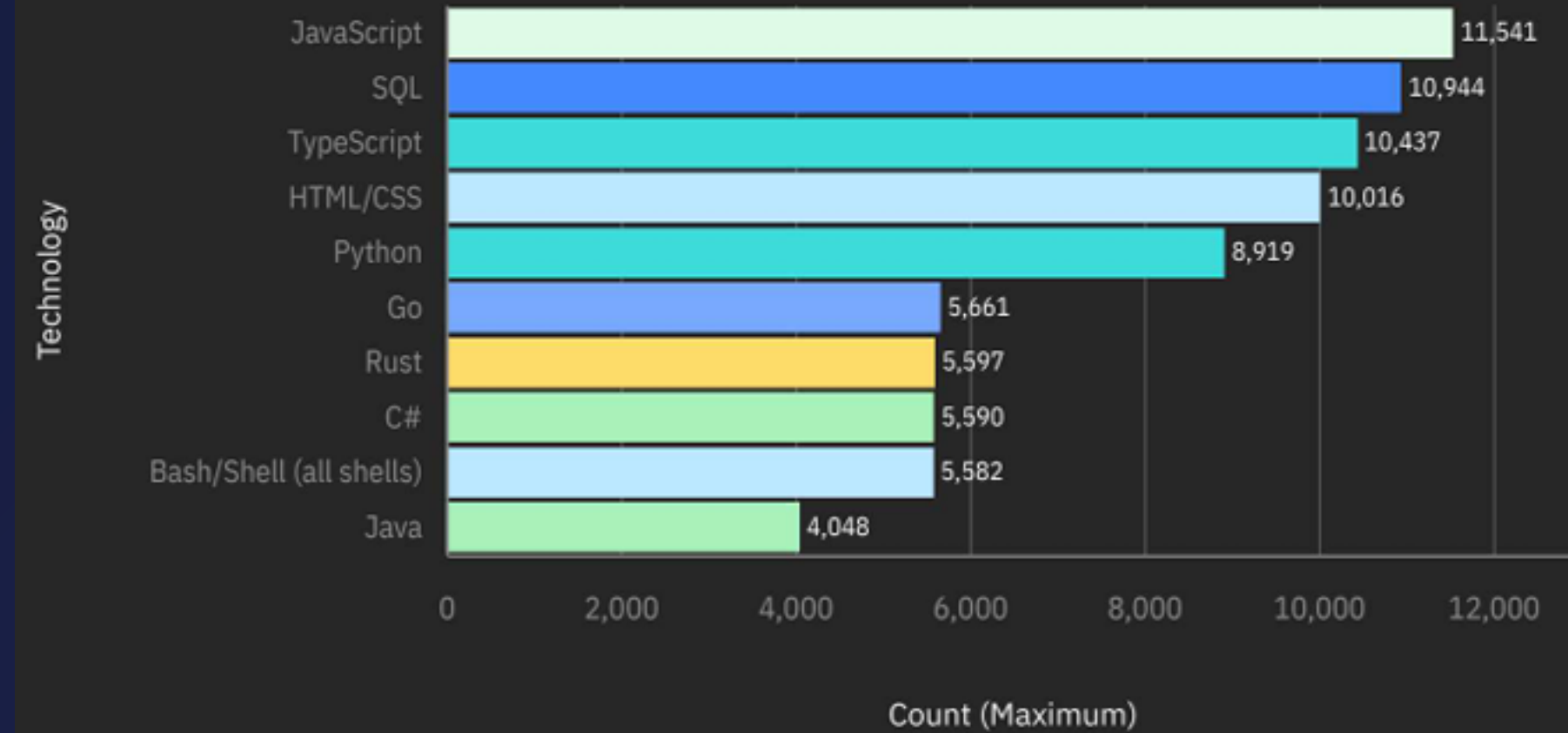
Current Year

Next Year

Top 10 Languages Worked With



Top 10 Desired Languages



Current Year

Findings & Implications

Database Trends

Findings

- JavaScript is the most used language, driven by web development demand.
- Python and SQL remain highly popular, reflecting growth in data and analytics.
- TypeScript and Rust are gaining interest, showing demand for modern technologies.
- Most desired languages mirror current usage, indicating stable market trends.

Implications

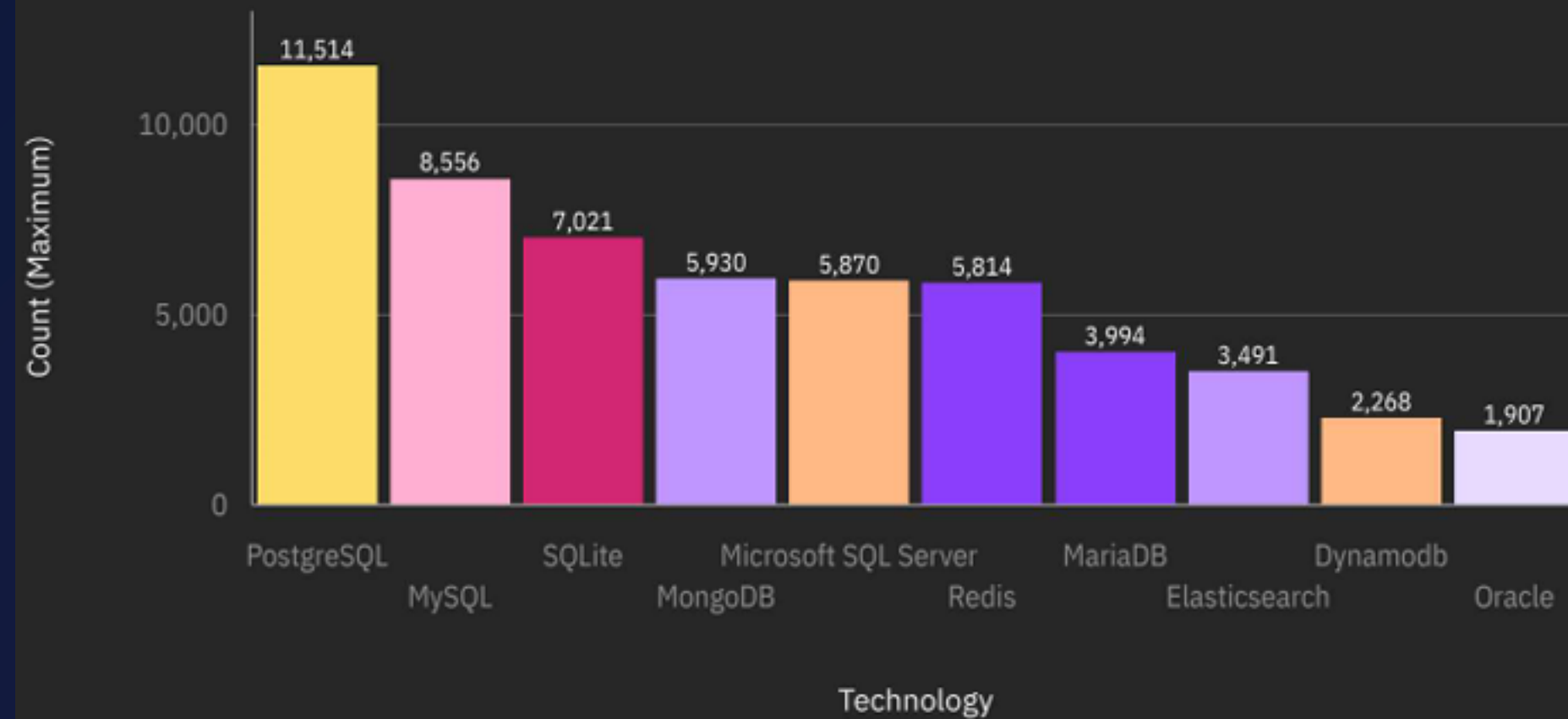
- Focus on JavaScript, Python, and SQL for strong career prospects.
- Learning TypeScript and Rust can provide a competitive edge.
- Companies should balance established and emerging technologies.
- Continuous upskilling is necessary to stay relevant.

Current Year

Database Trends

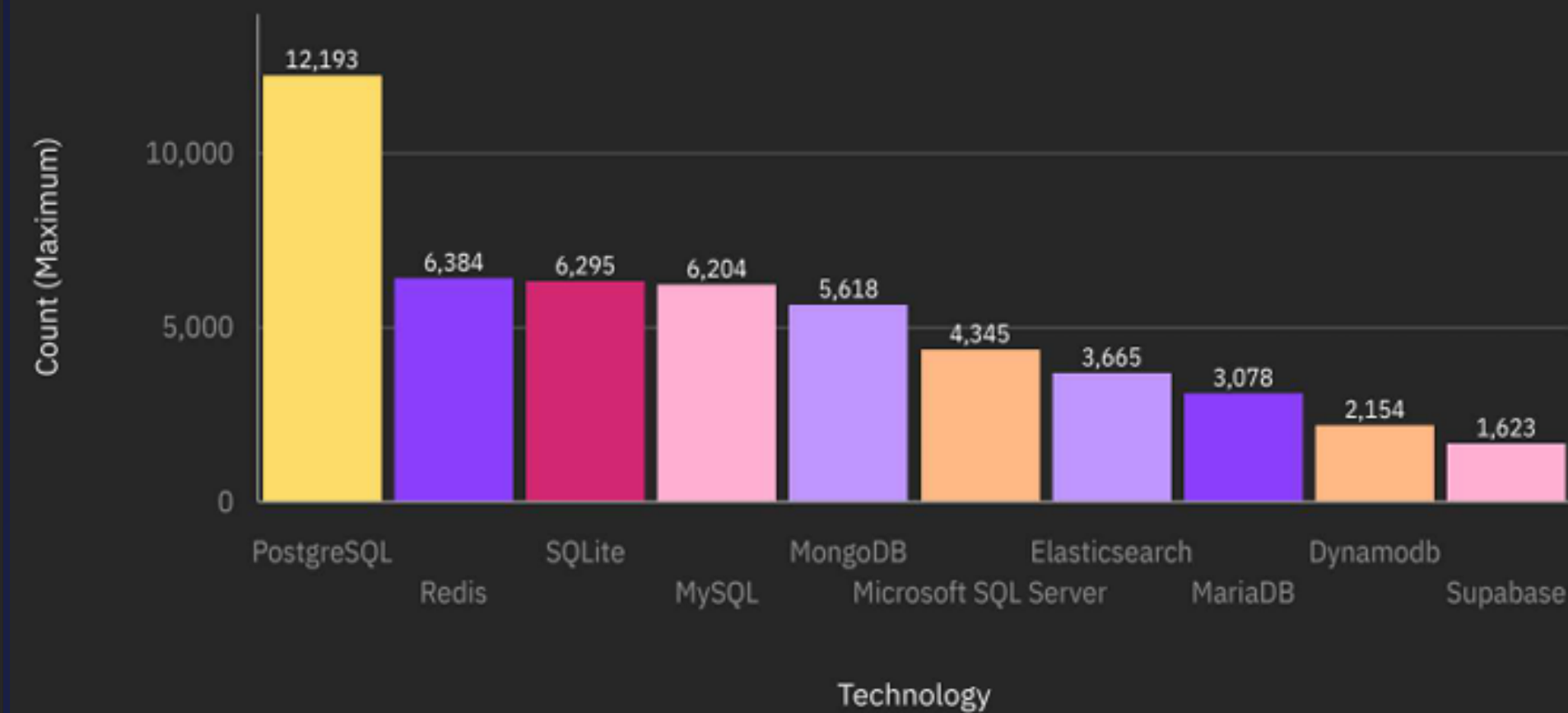
Current Year

Top 10 Database Worked With



Next Year

Top 10 Desired Databases



Current Year

Findings & Implications

Database Trends

Findings

- PostgreSQL is the most used database, showing strong adoption in modern applications.
- MySQL and SQLite remain widely used, reflecting their reliability and ease of use.
- PostgreSQL is also the most desired database, indicating continued future growth.
- Cloud and NoSQL databases are gaining interest, suggesting a shift toward scalable data solutions.

Implications

- Learning PostgreSQL and MySQL provides strong career opportunities.
- Knowledge of cloud and NoSQL databases can offer a competitive advantage.
- Organizations should invest in scalable and modern database technologies.
- Database professionals need skills in both traditional and emerging data platforms.

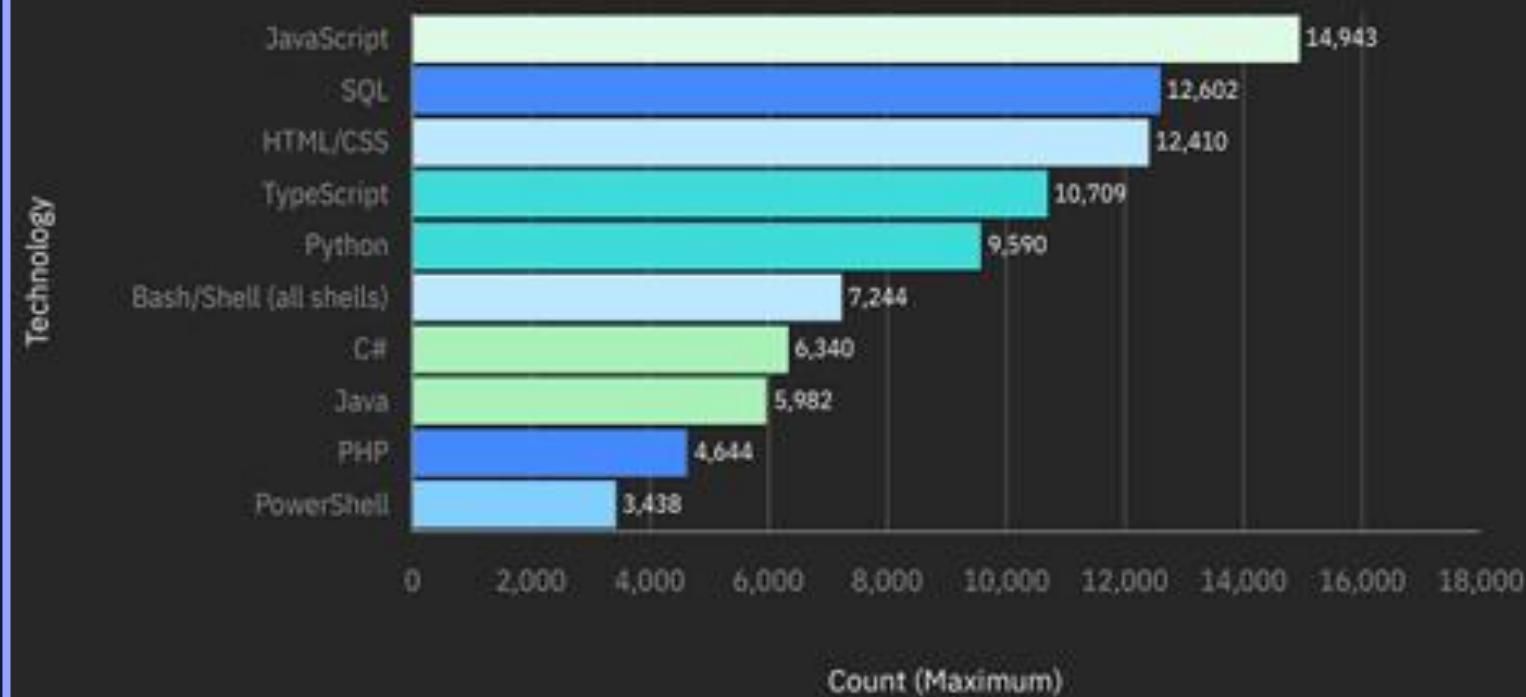
Current Year

Dashboard



Tab 1: Current Technology Trends

Top 10 Languages Worked With



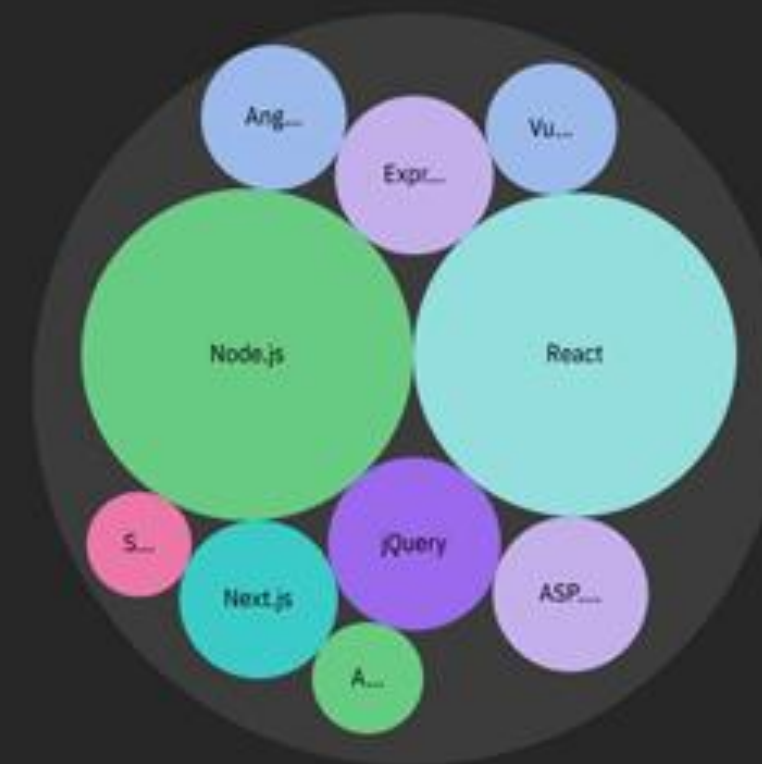
Top 10 Database Worked With



Top 10 Platforms Worked With

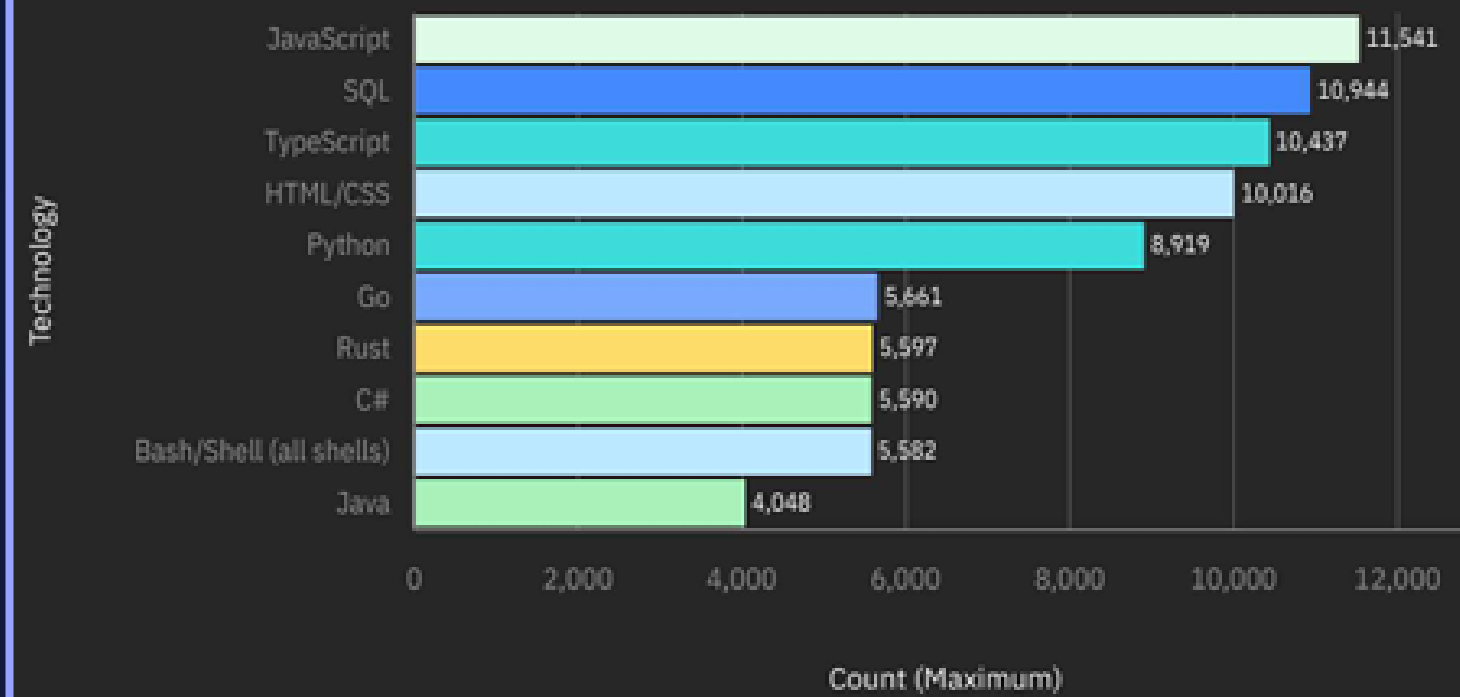


Top 10 Webframes Worked With



Tab 2: Future Technology Trends

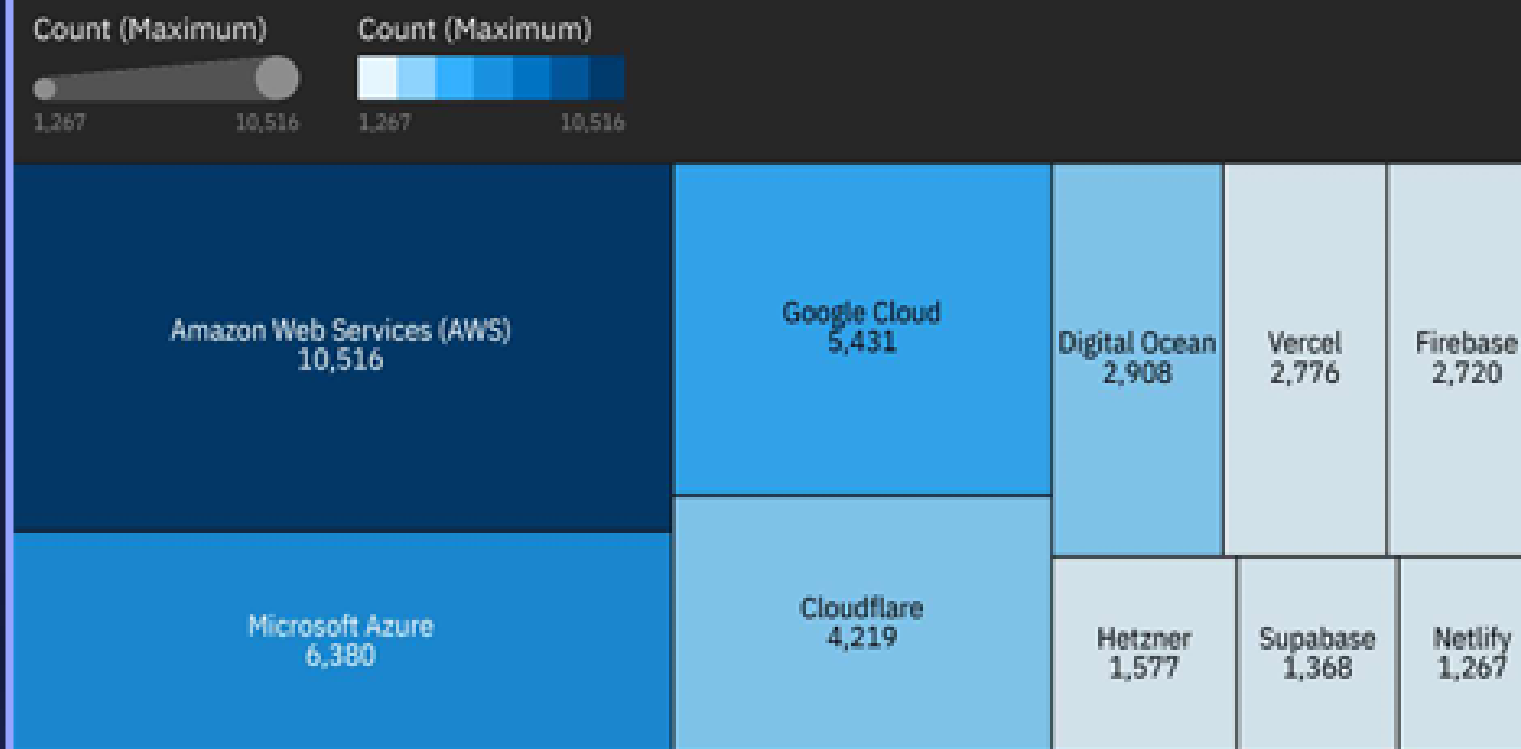
Top 10 Desired Languages



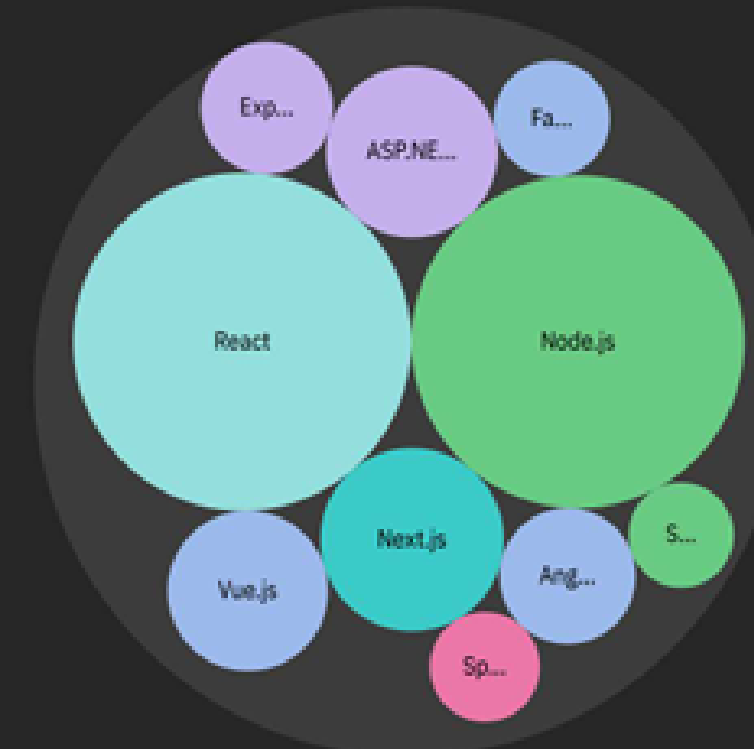
Top 10 Desired Databases



Top 10 Desired Platforms



Top 10 Desired Webframes

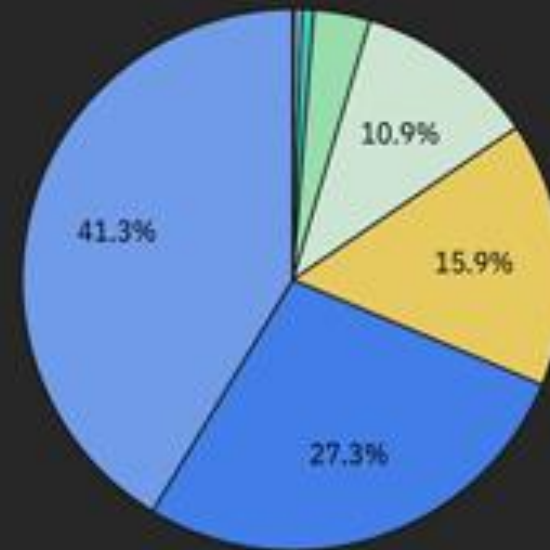


Tab 3: Demographics

Respondent Distribution by Age

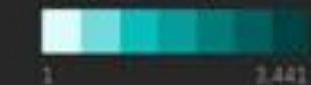
Age

- Prefer not to say
- 18-24 years old
- 65 years or older
- 35-44 years old
- Under 18 years old
- 55-64 years old
- 45-54 years old
- 25-34 years old

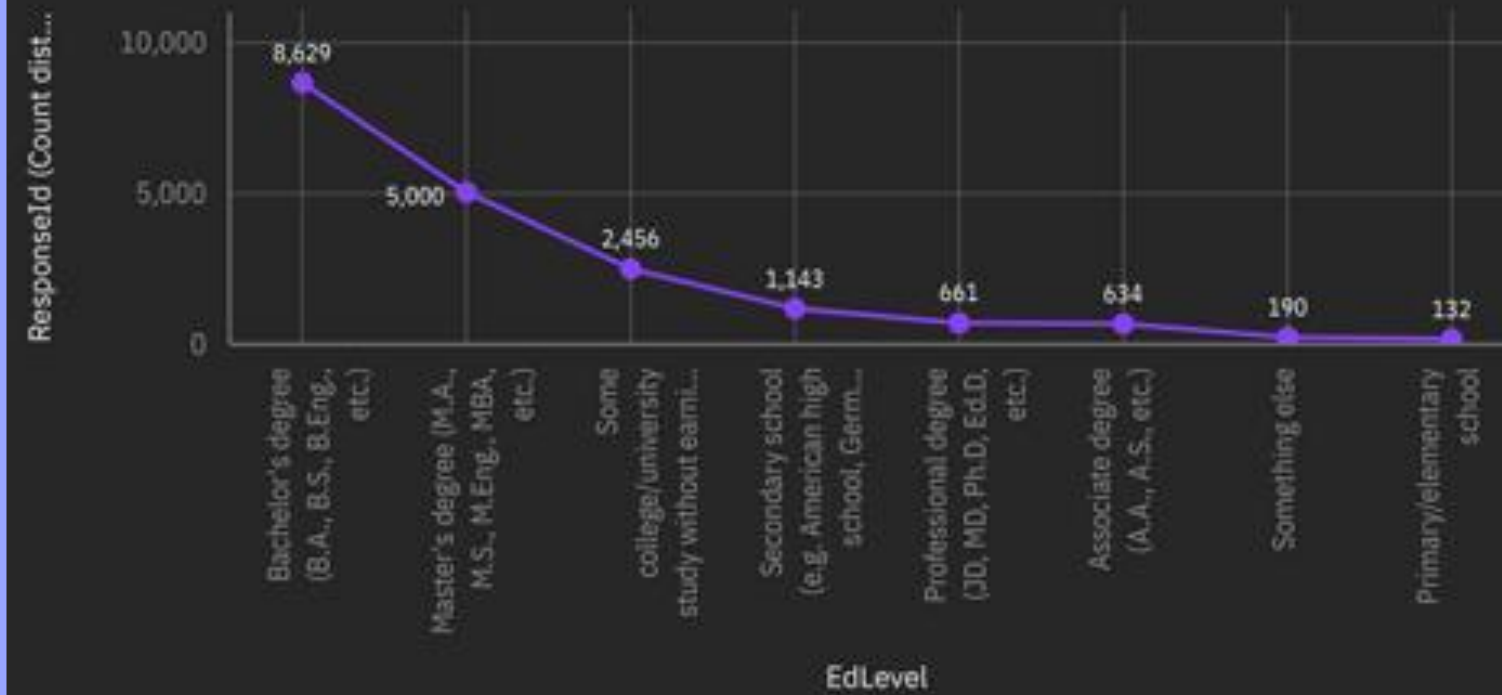


Respondent Count By Country

ResponseId (Coun...



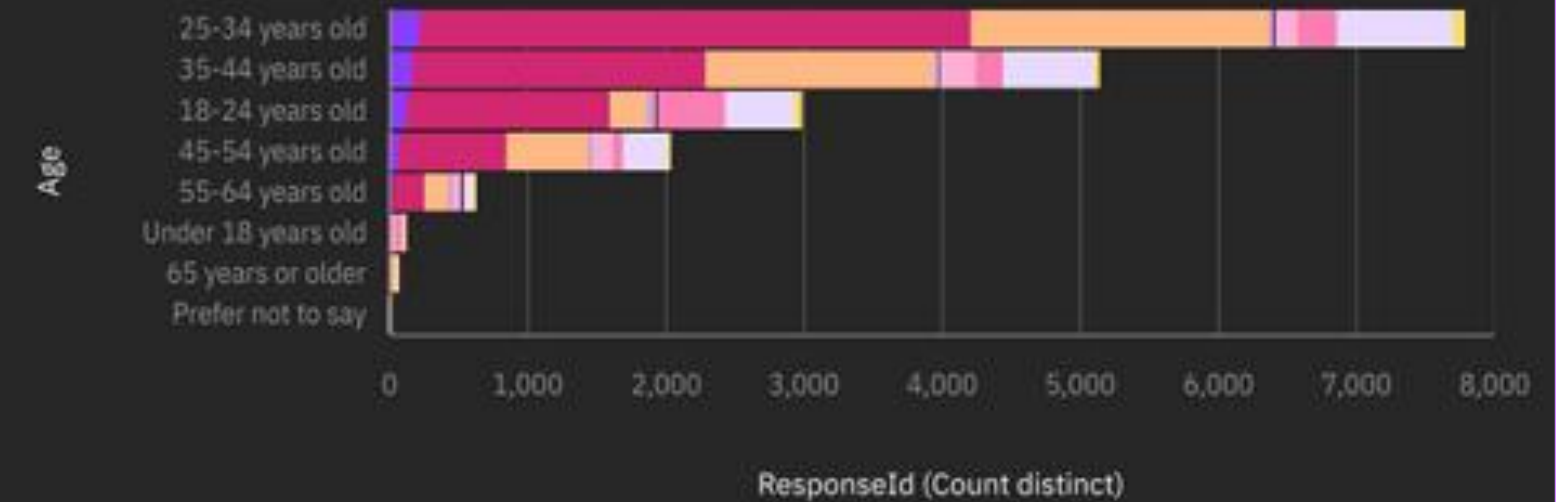
Respondent Distribution by Formal Education Level



Respondent Count By Age Classified By Education Level

EdLevel

- Associate degree (A.A., A.S., etc.)
- Primary/elementary school
- Some college/university study without earning a degree
- Bachelor's degree (B.A., B.S., B.Eng., etc.)
- Professional degree (JD, MD, Ph.D., etc.)
- Something else
- Master's degree (M.A., M.S., M.Eng., etc.)
- Secondary school (e.g. American high school, German gymnasium, etc.)



Discussion *Dashboard*

The dashboard highlights that the technology industry is largely driven by JavaScript, Python, SQL, and PostgreSQL, with strong and continuing demand for data, web, and cloud technologies. Current usage and future preferences are closely aligned, suggesting that these technologies will remain highly relevant while modern technologies such as TypeScript and Rust continue to gain momentum.

Current Year

Findings & Implications *Dashboard*

Findings

- Demand remains concentrated around few core technologies.
- Data & cloud technologies continue to gain importance.
- Interest in non relational databases and modern tools is increasing.

Implications

- Prioritize skills in data, cloud & modern development tools.
- Organizations should balance traditional and emerging technologies.
- Continuous learning remains essential in a changing tech landscape.

Current Year

Conclusion

- The tech industry is evolving rapidly with developer preferences shifting toward scalable cloud technologies
- Data-driven insights are becoming more essential for organizations to make informed decisions.
- Staying ahead requires continuous learning & adaptability as the developer landscape continues to transform

Current Year